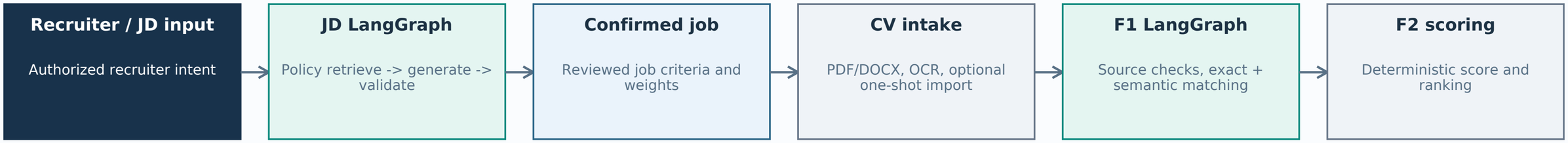
TalentTrace AI - final LangGraph workflow map

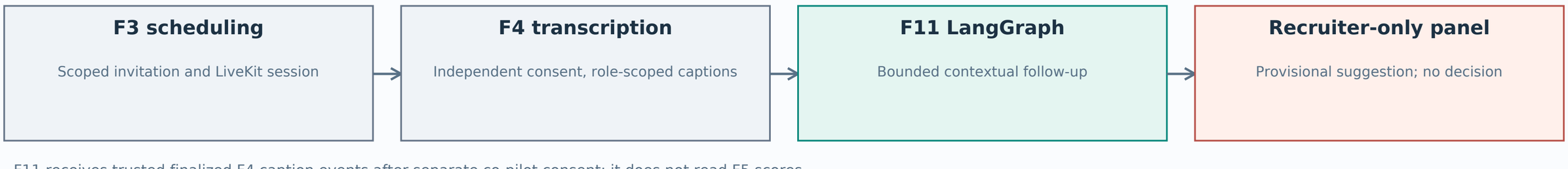
Architecture overview: the four implemented StateGraphs and their surrounding application services

01 - BEFORE THE INTERVIEW



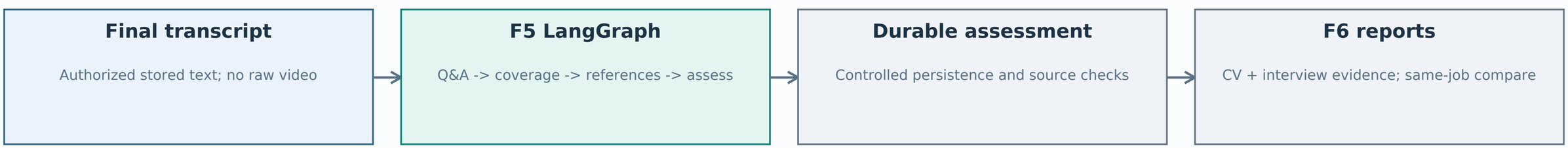
Downstream candidates must be eligible and have valid persisted assessment evidence before interview selection.

02 - DURING THE INTERVIEW



F11 receives trusted finalized F4 caption events after separate co-pilot consent; it does not read F5 scores.

03 - AFTER THE INTERVIEW



IMPLEMENTATION BOUNDARY

Only JD, F1, F5 and F11 are LangGraph StateGraphs. F2/F3/F4/F6, OCR, imports and job publishing are independent services. The diagram shows data dependencies, not a single continuously successful end-to-end test.

Workflow 1 - Job Description generation

Actual graph: backend/app/agents/jd/workflow.py / JobDescriptionWorkflow

IMPLEMENTED LINEAR STATEGRAPH



WHAT HAPPENS AFTER GRAPH COMPLETION?

Recruiter review and saved job

The recruiter can refine, save and confirm criteria for screening. These are API/UI services, not additional LangGraph nodes.

Controlled failure state

Missing configuration, policy retrieval failure, invalid citations or LLM failures produce sanitized errors, not approved JDs.

GRAPH STATE AND BOUNDARY

State includes

Request, retrieved policy chunks, generated draft, provider/model and validated response.

Checkpointing

The actual JobDescriptionWorkflow compiles with checkpointer=False; saved jobs are persisted outside the graph.

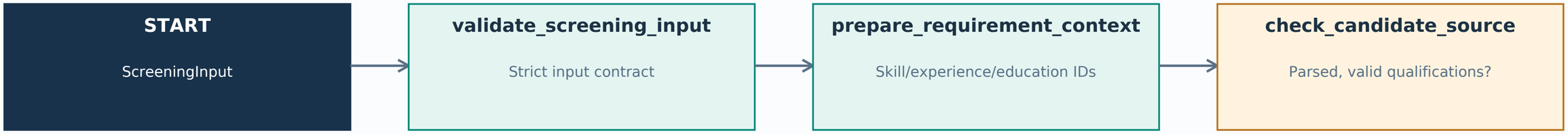
IMPLEMENTATION BOUNDARY

No automatic criteria approval and no hidden publishing graph. The optional publishing workflow is a separate service and requires user review before an external action.

Workflow 2A - F1 screening: preparation and matching

Actual graph: backend/app/agents/screening/workflow.py / CandidateScreeningWorkflow

PREPARATION AND SOURCE GATE



THREE POSSIBLE PATHS FROM THE SOURCE GATE

Prepare operation

build_ready_result -> END
Only source readiness was requested.

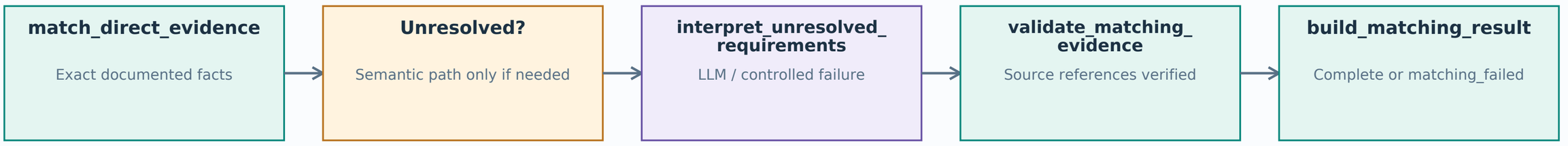
Source unavailable

build_source_unavailable_result
Screen -> no-decision; other modes -> END.

Ready for matching

Run direct matching and optional
LLM matching for unresolved criteria.

MATCHING SUBGRAPH



If no unresolved criteria: skip LLM and go directly from matching to evidence validation.

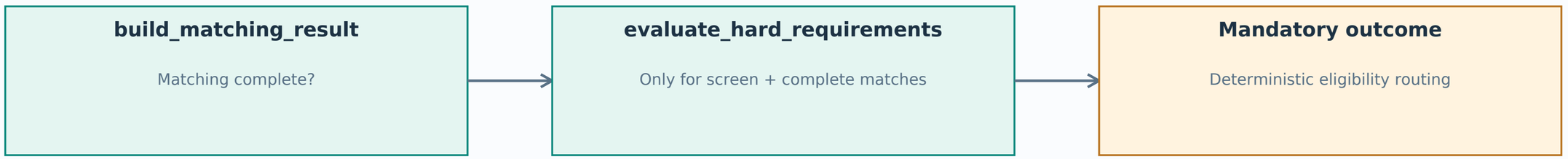
IMPLEMENTATION BOUNDARY

The graph has separate prepare, match and screen operations. Invalid source evidence or a provider failure results in explicit unavailable/matching_failed/no-decision states rather than invented eligibility.

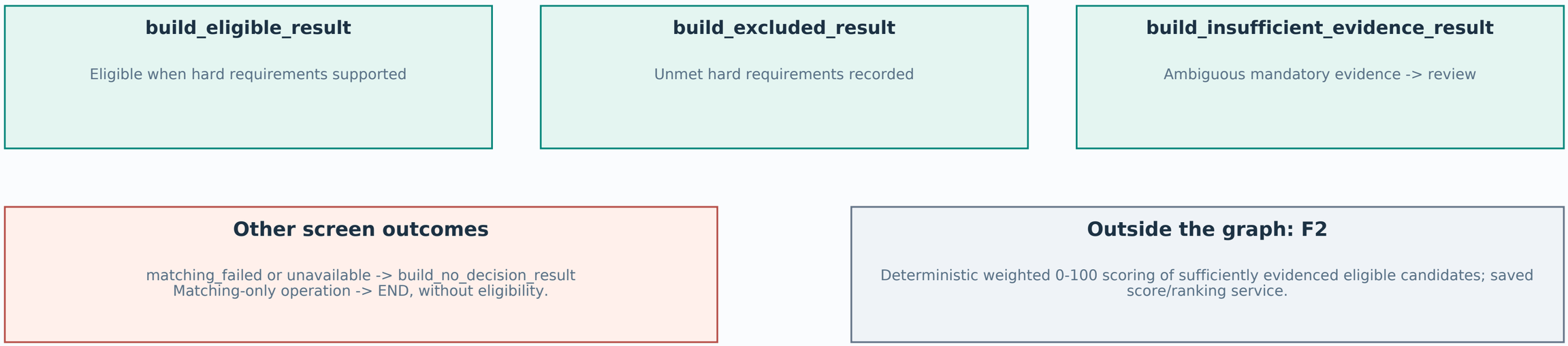
Workflow 2B - F1 decision routing, F2 and checkpointing

Screening decision branch and service boundaries / same graph definition as page 3

MANDATORY-REQUIREMENT DECISION BRANCH



CONDITIONAL RESULT NODES



RESUMABILITY BOUNDARY

An internal start_checkpointed_screening/resume_checkpointed_screening path can supply PostgreSQL AsyncPostgresSaver. It requires a matching thread and original request; the normal graph does not automatically use a saver.

IMPLEMENTATION BOUNDARY

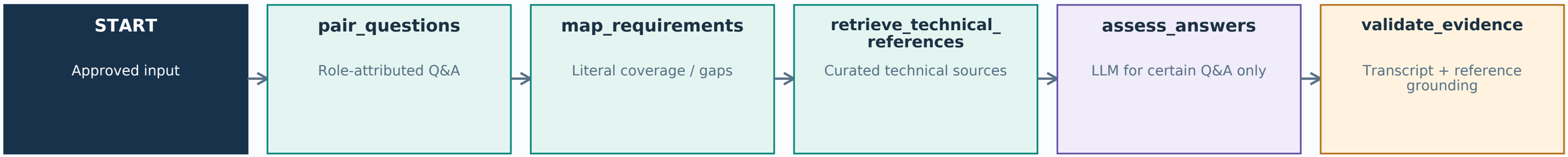
F1 is the LangGraph. F2 is regular deterministic code, not an LLM or a scoring graph. Incomplete, excluded or uncertain outcomes do not receive fabricated numeric scores. General cross-process live screening recovery has not been fully accepted.

Workflow 3 - F5 post-interview evaluation

Actual graph: backend/app/agents/evaluation/workflow.py / InterviewEvaluationWorkflow

AUTHORIZED INPUT OUTSIDE THE GRAPH

Final transcript, consent/retention checks and confirmed job revision are verified before START.



VALIDATION BRANCHES - ONE REPAIR MAXIMUM

VALID

Return validated advisory assessment
-> durable manager saves result.

REPAIR ONCE

Retry assess_answers after fixed grounding feedback.
Validate the revised result again.

FAIL CLOSED

If still invalid, raise controlled AssessmentError.
Never fabricate a source citation.

WHAT THE GRAPH DOES NOT DO

Uncertain Q&A

Ambiguous question-answer pairing is marked not_assessed; the LLM is not asked to infer competence for that pair.

Persistence

The graph itself uses checkpointer=False. InterviewAnalysis manager and PostgreSQL records live outside the graph.

No combined score

Relevance, technical accuracy, completeness and articulation remain evidence-linked advisory statuses. F6 does not invent a unified score.

IMPLEMENTATION BOUNDARY

The current technical corpus is curated documentation with local lexical/alias matching; optional local embedding retrieval is distinct. Separate historical real-model success and current synthetic browser tests do not establish full live F4-to-F5 acceptance.

Workflow 4 - F11 live interviewer co-pilot

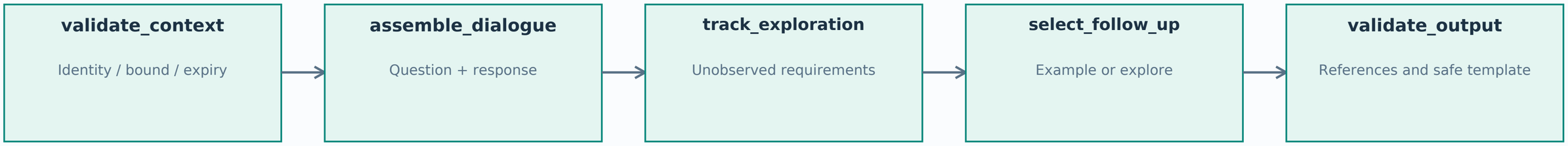
Actual graph: backend/app/agents/copilot/workflow.py / build_graph

TRUSTED INPUT GATE - OUTSIDE THE GRAPH

F4 finalized caption events + approved criteria + two-party F11 consent + current run/caption IDs

No public browser text-ingestion endpoint and no separate LLM provider.

MODEL-FREE LANGGRAPH FLOW



CONDITIONAL OUTPUT AND DELIVERY BOUNDARIES

SUGGESTED

Validated provisional follow-up.
No ability, personality, score or hire/reject judgment.

WAIT / ABSTAIN / REJECT

Incomplete dialogue, ambiguous evidence, expiry, unsafe output or invalid event identity.

RECRUITER-ONLY DELIVERY

API/status and React co-pilot panel.
Cooldown, dismiss, withdrawal and TTL outside graph.

IMPLEMENTATION BOUNDARY

The co-pilot LangGraph uses checkpointer=False and bounded in-memory state, not a model. Default runtime gates are off. Offline browser consent, private delivery and suggestion tests passed; full real Cloud speech-to-F11 acceptance is still partial.